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DATA SCIENCE INSIGHTS FOR SOCIETY

GEO

Data Preprocessing and Potential Changes Detection in Soil Surface Roughness of Crop Fields with Synthetic Aperture Radar (SAR)

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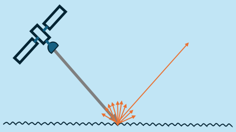
INTRODUCTION

Synthetic Aperture Radar (SAR) is a remote sensing technology that transmits signals to a surface and records the bounced back signal and time delay. Soil surface roughness is one of the main factors affecting the reflection of SAR backscatter, and changes in soil surface also cause changes in SAR.

In this work, we want to **use SAR to detect changes in surface roughness of agricultural fields due to farming activities.** By **comparing changes in satellite visit-to-visit SAR of parcels in the same area, we can distinguish changes due to weather, which affects all fields in the same manner, from the changes caused by farming, which can cause outlying changes in the worked fields.**



Rough Surface

Backscatter is the reflected signal perceived at the sattelite.

DATA PREPARATION

Field Boundary

The area of the study is in **Flevoland, Netherlands.**
The **satellite images** were query from **Google Map API.**
Fields' boundary was annotated with **QGIS.** There are 196 parcels in total.

https://qgis.org/
https://www.google.com/maps

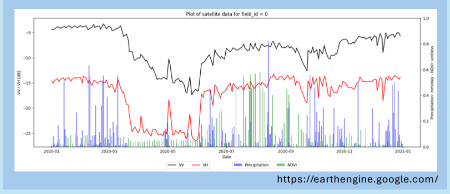
Farm Activities

Farming activities was acquired from the farming log of Harrysfarm's. There were total of **102 activities during 2018 to 2022** including fertilize, harvest, plant, ridge, and tillage over **5 fields.**

https://www.harrysfarm.nl/

Satellite Data

Satellite data were queried from Google Earth Engine including **SAR in VV and VH polarisation**, and also Precipitation and NDVI data.
We can scale the signal into decibel and normalise the signal into 38 degree incidence angle.



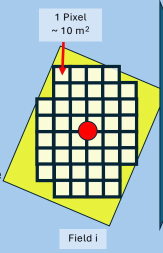
https://earthengine.google.com/

METHODS

SAR Data

SAR data is presented as 2D images (or matrices) overlaying the measured area. Each pixel contains a backscatter signal value at a specific time point.

To represent a data point for a field at a single time point, we **calculated the average of all pixels** contained within each field boundary. This averaging approach helps smooth the signal and reduce noise.



Features

Satellite data were queried from Google Earth Engine, including **SAR in VV and VH polarizations**, as well as precipitation and NDVI data. The SAR signals were converted to decibels (dB) and normalized to a 38-degree incidence angle.

Value Differences	$[VV_t - VV_{t-1}, VH_t - VH_{t-1}]$
Value Ratio	$[\frac{VV_t}{VV_{t-1}}, \frac{VH_t}{VH_{t-1}}]$
Ratio and Difference of Ratio	$[\frac{VV_t}{VV_{t-1}}, \frac{VH_t}{VH_{t-1}}, \frac{VH_t}{VV_t} / \frac{VH_{t-1}}{VV_{t-1}}]$

Approach

To detect changes in soil surface roughness, we used farming activities as a proxy. We assumed that farming activities would cause changes in the surface that would be detectable via changes in SAR. Additionally, these changes would cause outlying differences in the signal compared to other fields in nearby areas.

We tested our hypothesis **using farming activities from Harrysfarm**. If the outlier detectors flagged a parcel in Harrysfarm as an outlier during the same period that a farming activity took place in that parcel, we counted it as a true positive; if not, it was counted as a false positive. A false negative occurred when the detector failed to identify a parcel as an outlier despite farming activity taking place.

Experiment Setup

We split the data into **training period** and **testing period.**

Training Period (2017-2020)	Testing Period (2021-2022)

To detect farming activities for each time point, we used outlier detection methods to identify outliers. **Three outliers detection methods were tested.**

DBSCAN	Isolation Forest	Mahalanobis Distance with χ^2

We used grid search to find the best parameters in the training period. Then, we tested for the final performance on the testing period. The evaluation metric was the F1 Score.

$$F1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall}$$

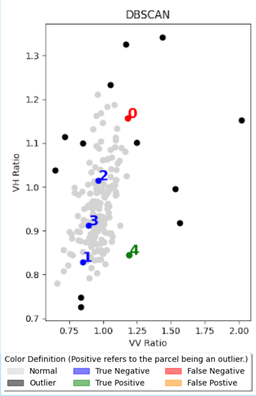
RESULTS

The results in Table A showed that DBSCAN performed the best with an F-Score of 0.3265. Isolation Forest and Mahalanobis Distance with χ^2 performed similarly well. For illustration, Figure B shows an example detection using DBSCAN. In this figure, we can see that parcel 4 is a true positive. Parcels 1, 2, and 3 are true negatives, while parcel 0 was a false negative.

A. Test Set Result of each outlier detection models using the best parameters

Model	Feature	Preprocessing	Best Parameters	F-Score
DBSCAN	[VV Ratio, VH Ratio]	Decibel (by orbit & 38° Norm)	`eps=0.5` `min_samples=3` `metric="euclidean"` `leaf_size=10`	**0.3265**
Isolation Forest	[VV Ratio, VH Ratio]	Decibel (by orbit & 38° Norm)	`n_estimators=50` `contamination=0.1`	0.2395
Mahalanobis Distance with χ^2	[VV Difference, VH Difference]	Decibel (by orbit)	`alpha=0.01`	0.2346

B. Example detection for period from 2022-09-12 to 2022-09-23



For comparison, we can compare the **F-Score of the training set** with **Huber et al. 2023 at 0.55** with **our DBSCAN that achieved 0.2243.** The result may fall short from the previous work, however, our method can scale easier as the method are simple to calculate.

Reference
[1] Zhu, L., Walker, J.P., Ye, N., Rüdiger, C., 2019. Roughness and vegetation change detection: A pre-processing for soil moisture retrieval from multi-temporal SAR imagery. Remote Sensing of Environment 225, 93–106. https://doi.org/10.1016/j.rse.2019.02.027
[2] Huber, M., Kumar, V., Steele-Dunne, S.C., Rommen, B., 2023. Sentinel-1 InSAR Coherence as an Indicator of Monitor Farming Activities, in: IGARSS 2023 - 2023 IEEE International Geoscience and Remote Sensing Symposium. Presented at the IGARSS 2023 - 2023 IEEE International Geoscience and Remote Sensing Symposium, IEEE, Pasadena, CA, USA, pp. 429–432. https://doi.org/10.1109/IGARSS52108.2023.10281522
[3] Schreuder, H. Crop Growth 2018-2022. https://www.harrysfarm.nl/. Accessed on 18 July 2024.